

Design and Analysis of Algorithms

Course Information

- 1 Why Study Algorithms?
- 2 How to Study Algorithms?

Basic Course Information

课程名称: 算法设计与分析

课程号: sd046301400

- 1-16 周 (每周) 会文北楼 244 周三上午: 1-2 (8:00-9:50)
- 1-16 周 (单周) 会文南楼 225 周四上午: 1-2 (8:00-9:50)

1 Why Study Algorithms?

2 How to Study Algorithms?

A composite image of Earth from space. The Earth's horizon is visible, showing a bright sun on the left side, creating a lens flare effect. The atmosphere is glowing with a blue and orange hue. The background is a dark, starry space.

Just one simple idea...

can change the world.

- Rana Currie -

Image: Pixabay.com



Two ideas that changes the world!

Typography

1448, German, Johann Guternberg used movable metal type to print the Latin Bible



Typography accelerates the spread of knowledge and triggers a chain reaction:

- literacy spread $\Rightarrow$ Dark Ages ended $\Rightarrow$ human intellect was liberated $\Rightarrow$ science and technology triumphed $\Rightarrow$ industrial revolution happened

Imagine a world in which only an elite few could read and write, and knowledge can not be rapidly accumulated or disseminated.

Other Voice



But others argue that the key innovation was not typography, but the *algorithm*

Algorithm

Origins: decimal numeral system (thought to be natural in hindsight)

- 10 symbols $\Rightarrow$ even large numbers can be expressed compactly (invented in India around AD 600)
- basic methods for add, mul, div, even square roots and π (9th century, Arabic, Baghdad, Al-Khwarizmi)



Key Features of Decimal System

- procedures are precise, unambiguous, mechanical, efficient, correct $\leadsto$ Algorithms (有效的计算)

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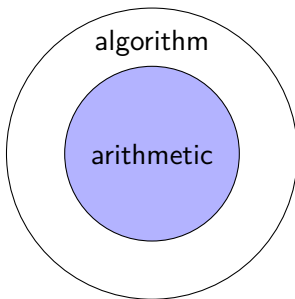
Back to 1448: imaging add/mul two Roman numbers: MCDXLVIII+DCCCXII.
Counting on your fingers would not be enough.

Algorithm Etymology

Spread to Europe around 12th century $\leadsto$ plays an enormous role in Western civilization (science and technology, commerce and industry)

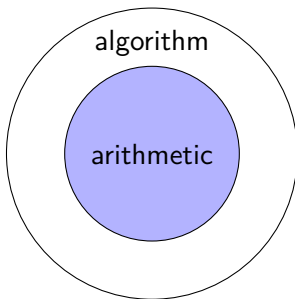
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Computer Era

- traditional algorithms shrinks to arithmetic
- modern algorithms evolve from positional system (位值系统) + arithmetic unit $\leadsto$ scientists develop algorithms for all kinds of problems $\leadsto$ ultimately change the world

Why Study Algorithms

Internet. Web search, packet routing, distributed file sharing, ...

Computer graphics. movies, video games, virtual reality, ...

Multimedia. MP3, JPG, DivX, HDTV ...

Artificial Intelligence. face recognition, PS, more AI algorithms

Social networks. recommendations, advertisements, ...

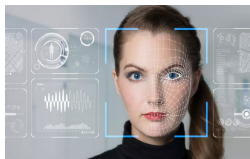
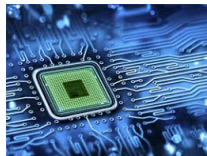
Computers. circuit layout, databases, caching, compilers, ...

Biology. human genome project, vaccine manufacture, ...

Physics. N -body simulation, particle collision simulation, ...

Importance: Look Around You

Google
YAHOO!
bing



Algorithms are both interesting and useful.
We live in the world shaped by algorithms!

A Detour

What is the connection between algorithm and cryptography?

Cryptographic Algorithms

Algorithms generally aim to solve problems correctly and efficiently

- make us live in a better world (compare 12306 now and 15 years ago)

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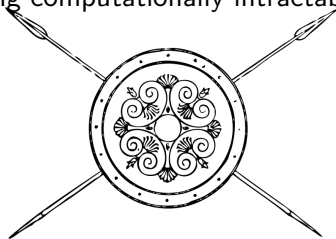
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Cryptography is **Algorithms** in the **area of information security**

- honest parties can perform the intended operations efficiently
- malicious adversaries cannot violate the security guarantees (solve specified task efficiently) without solving computationally intractable problems



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Algorithm design and analysis is the cornerstone of CS

- widespread applications
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- algorithm design: 11
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- cryptography: ≥ 12

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Many problems remain open

$\mathcal{P} \stackrel{?}{=} \mathcal{NP}$ is one of the most important questions in this century

Expect you can solve it in the future.

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Preliminary about algorithms

- mathematical background
- data structure

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Algorithm design paradigm and analysis methods

- divide-and-conquer
- greedy strategy
- dynamic programming
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Advanced topics

- complexity theory
- randomized algorithms

Topics Not Covered In This Course

Linear programming and reductions

- bipartite matching
- network flow

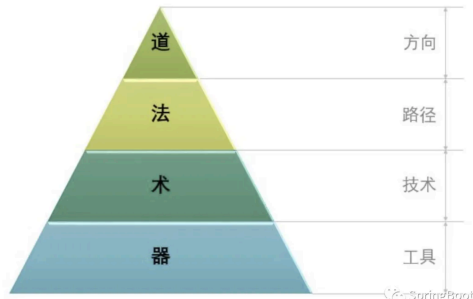
Quantum algorithms

Advanced data structures

- segment tree

The Essence of University Education

- Cultivate long-term attributes (道: 本质)
- Form short-term capability (法: 方法)
- Master special skills (术: 技术)
- Learn universal knowledge (器: 工具)



Detailed Learning Objectives (器、術)

Algorithm design: Master systematic **problem-solving** methods

- ① abstract and formalize a problem
- ② design a correct and efficient algorithm
- ③ prove the algorithm's correctness

Algorithm analysis: Develop **rigorous analysis** skills

- know how to evaluate the performance of algorithms
-

Tips

- theory: think rigorously and keep asking yourself why
- practice: implement algorithms using your favorite programming languages

Broader Learning Objectives (法、道)

Develop **critical thinking** (批判性思维)

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The benefits of critical thinking

- Individual level: Thinking is productivity.
- Interpersonal level: Defend against bullying and mind manipulation.
- Societal level: Not be dominated by advertising or conventional wisdom.

Course Website

<https://yuchen1024.github.io/teaching/algorithm/>

Syllabus

Assignments

- electronic submission
- graded for correctness, clarity, conciseness, rigor, and efficiency
- recommendation: using $\text{\LaTeX}$ template for writing solutions
- no collaboration, no Google

Lecture slides

...

总成绩 = $0.1 \times$ 平时成绩 + $0.1 \times$ 编程实践 + $0.3 \times$ 课后作业 + $0.5 \times$ 考试成绩

References and Resources

Online resources

- leetcode
- online judging system: ZOJ, POJ

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Textbooks

- Algorithms. Sanjoy Dasgupta, Christos Papadimitriou, and Umesh Vazirani.
- 算法设计与分析 (第二版). 屈婉玲, 刘田, 张立昂, 王捍贫.



Figure: 屈婉玲

<https://zhuanlan.zhihu.com/p/193792826>